

"ContextGenOS"

A Universal Personal Memory & Context Layer

The Core Idea

An open source, **locally-run personal AI memory system** that sits as a middleware layer between *you* and *any* AI model or app.

It captures, structures, and selectively injects your personal context (work, habits, preferences, relationships, projects) into any AI interaction without sending your data to any cloud.

Why It Doesn't Exist Yet (Properly)

- **Mem.ai** is closed source and cloud-dependent
- **Obsidian + AI plugins** are fragmented and require heavy manual curation
- **OpenAI Memory** is siloed to ChatGPT only
- No project owns the *cross-model, privacy-first, programmable* personal context layer

What Makes It Succeed

Technical hook: A local vector + graph hybrid store with a universal context API that any app or model can call via a simple SDK or MCP server. Works with Claude, GPT, Gemini, local LLMs, anything.

Privacy moat: 100% local-first by default. This is the single biggest unmet demand in AI right now. People want AI that knows them without surveillance.

Contributor magnet: Modular "context collectors" — plugins for email, calendar, GitHub, browser history, notes, health data. Every developer has a domain they'd love to contribute.

User magnet: The value compounds daily. The longer you use it, the smarter every AI tool you touch becomes. That's a powerful retention loop.

Architecture Sketch

[Data Sources] → [Context Collectors] → [ContextGenOS Core]



[Universal Context API / MCP Server]



[Claude / GPT / local LLM / any AI app]

Core stack: **Rust or Go** for performance, **DuckDB + Qdrant** for local hybrid search, **MCP**

protocol for AI model integration, **Python SDK** for contributor accessibility.

Why Now

- MCP (Model Context Protocol) just became a real standard — the infrastructure moment has arrived
- Local hardware (Apple Silicon, consumer GPUs) is now powerful enough for always-on local AI middleware
- Growing regulatory pressure (GDPR, EU AI Act) is making cloud-dependent personal AI politically fragile
 - No dominant open source project owns this space yet

Comparable Successors to Study

- **Ollama** — succeeded by making local AI *deadsimple* → ContextGenOS should do the same for personal context
- **LangChain** — succeeded by being the glue layer → ContextGenOS is the *personal data* glue layer
- **Home Assistant** — succeeded with local-first, plugin ecosystem → same playbook, different domain

This sits at the intersection of **privacy**, **developer tooling**, and **daily AI utility**, three of the strongest forces driving open source contribution right now. The team that ships a clean v0.1 with MCP support and 3–4 context collectors (browser, calendar, notes) will own the category.